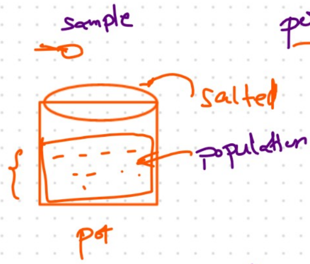
Statistics

def: It is a domain of mathematics & science where we focus on collecting, organizing, analyzing, and interpreting data in order to make decisions

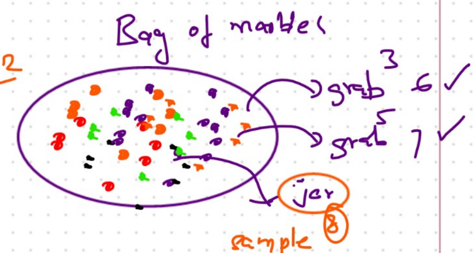
real world: answers questions using data & numbers when we cannot observe everything directly

What problem does Statics Solves ? : express answers not just yes / No rather as numbers, probabilities and percentages

Analogy ①



Analogy 2



* measure : things we can't fully observe
* Estimate: the unknown from the known
* Answer: question like
  1) what is the average
  2) what is the chance
  3) if x happens, does y follows

not guess ↓ backed by real number & probabilities

finally: We need to defend decisions with evidence

Core Concepts Breakdown

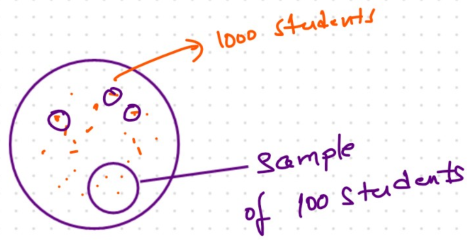
① Population: is about all the data — every single item, person, or entity you are interested in

→ We almost never work directly with the population because

* too large
* too expensive
* impossible to collect

** * population: → All the data not part of it

② **Sample:** A sample is a small part of population



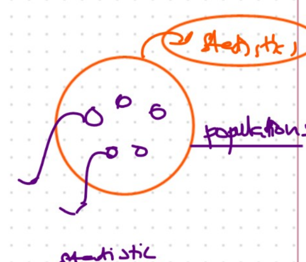
* Sample quality determines the quantity of your statistical conclusion

Important points: Bigger the sample ⟶ Better the estimate

③ **Parameter** It is any descriptive measure of population

mean, median, variance, percentage, count, max, min . . . . . .

Parameter = describe Population



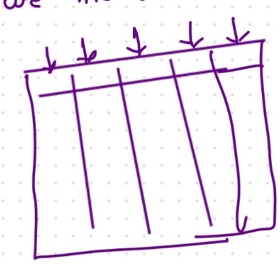
④ **Statistic** : It is a descriptive measure of a sample

Statistic = describes sample

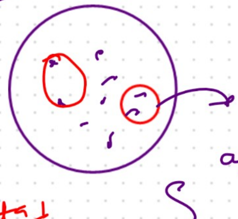
⑤ **Variable** : It is Something we measure or observe ⟶ column names ⟶ variable

not actual value

Dataframe



Exit poll



age group | cated ⟶ Student → height, weight, name, age, marks

Employee → height, weight, name, age, salary, emp

18 | P1 | P1

19 | P2

24 | P3 | P2

P3

Party 1 18 – 20

Party 20 – 25

25 – 35 Party

25 – 30

28 – 30

⑥ Observation: It is one data point — for a variable

⑦ Data: It is a collection of observation

⑧ Types of Variables

① Qualitative (categorical values)

cannot ask "how much ?"

ex: Gender, City, subject, Colour, Blood Group

② Quantitative (Numerical values)

You can ask "how much ?"

ex: Age, Height, weight, Temperature, Salary, battery %

Questions: A company has 5K employees, HR surveys 200 of them about job satisfaction. From the survey, they calculate the avg satisfaction score is 7.4/10

A Population, Sample, Parameter, Statistic and variable

Population → 5K

Sample → 200

Parameter → avg

Statistic → 7.4

variable → Job statisfaction score

Right !!!!

wrong ×

Ex2 A researcher wants to understand the avg monthly screen time of teenagers in India, she collects data from 1500 teenagers across 10 cities

A 5.2 hrs/day

① population approx 253 million (13-19)

② Sample 1500 teenagers

③ Is it realistic to study the population here ? impossible

④ The average is parameter or a statistic here

parameter ? !!!!

statistic ? ✓

⑤ what would make this estimate more reliable

biased → Not reliable

represent → all areas → rural

× different / more states

× different background

f_1 f_1 f_2

x_1 x_2 x_3 x_4 → y

Predict

Example

House price

~~fountain~~ pool

Inputs

x_1 x_2 x_3 x_4 x_5 y

No. of rooms | Sq. ft | Area | small | Build

→ 2 → 900 → Nagawara

3 1200 E-C

my Bay Mall